

Heilbronn's Triangle Problem: Ten Points

1 Definition

Let $U = [0, 1]^2$. For a finite set $P \subseteq U$, define

$$h(P) = \min_{\substack{x, y, z \in P \\ \text{distinct}}} \text{area}(\Delta(x, y, z)), \quad (1)$$

where $\Delta(x, y, z)$ denotes the triangle formed by the three points x, y, z , and degenerate triangles have area zero.

Definition 1 (The n th Heilbronn number). For $n \geq 3$, the n th *Heilbronn number* of the unit square is

$$H(n) = \max_{\substack{P \subseteq U \\ |P|=n}} h(P). \quad (2)$$

A set attaining this maximum is called an *optimal Heilbronn configuration*.

The maximum exists by compactness. The use of the unit square is convenient but not essential for asymptotic questions: for any two fixed convex bodies with nonempty interior, the corresponding Heilbronn numbers differ by multiplicative constants depending only on the two bodies.

2 Current best result

Optimal configurations have been determined or globally certified for $3 \leq n \leq 9$. The first unresolved case is $n = 10$. The best known configurations satisfy

$$H(10) \geq 0.04653741958254,$$

$$H(11) \geq \frac{1}{27},$$

$$H(12) \geq 0.03259885869182.$$

None of these values has been proved globally optimal.

3 problem formalization

Determine the exact value of $H(10)$ and classify all ten-point configurations in the unit square that attain it.